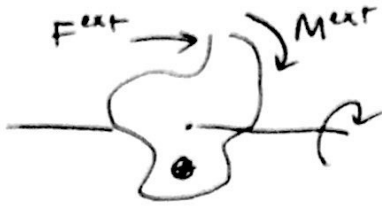


MAE 4730
Lecture 38
11/28/2018

Today:
- Dynamic balance review
- Euler equation

Balance



$\vec{F}^{ext}$ and $\vec{M}^{ext}$ rotate with the object
- Because they rotate w/ the object, they change in the fixed frame
* No vibration $\vec{F}^{ext} = 0$, $\vec{M}^{ext} = 0$

$$\text{LMB: } \vec{F}^{ext} = m\vec{a}_G = \dot{\vec{\omega}} \times \vec{\omega} \times \vec{r}_{G/O}$$

$$= \vec{0} \quad \underbrace{\text{only if } G \text{ is on axis}}_{\text{static balance}}$$

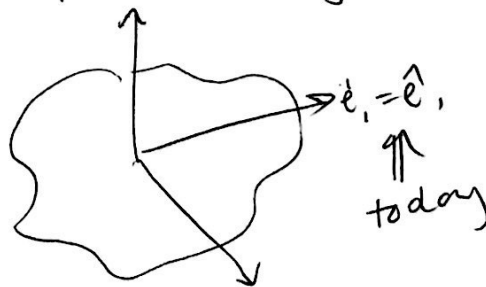
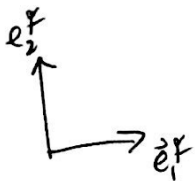
$$AMB/O = AMB/G$$

$$\vec{M}^{ext} = \vec{\omega} \times (\underline{I}_G \cdot \vec{\omega}) = 0 \quad \underbrace{(\text{only if } \vec{\omega} \text{ is principal axis } \underline{I})}_{\text{dynamic balance}}$$

$$\begin{bmatrix} m & 0 & x \\ 0 & m & \Delta \\ x & \Delta & m \end{bmatrix}$$

Any symmetric matrix will have 3 mutually orthogonal eigenvectors

Euler equation: stability of spinning about principal axis



$$\vec{\omega} = \vec{\omega}$$

$$\omega_i^f e_i^f = \omega_i e_i \rightarrow \text{today's notation}$$

Rigid object AMB

$$\vec{M}^{ext} = \dot{\vec{H}}_G = \underline{I} \cdot \dot{\vec{\omega}} + \vec{\omega} \times (\underline{I} \cdot \vec{\omega}) = 0$$

$\uparrow$ torque free motion

$$\dot{\vec{\omega}} = \underline{I}^{-1}(\vec{\omega} \times (\underline{I} \cdot \vec{\omega})) \quad (*)$$

$$\underline{\omega} = \omega_i e_i \quad \underline{I} = I_1 e_1 e_1 + I_2 e_2 e_2 + I_3 e_3 e_3$$

$$[I]_e = \begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix}$$

Matrix form of (*)

$$\begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix} \begin{bmatrix} \dot{\omega}_1 \\ \dot{\omega}_2 \\ \dot{\omega}_3 \end{bmatrix} = - \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} \times \begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix}$$

$$\begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix} \begin{bmatrix} \dot{\omega}_1 \\ \dot{\omega}_2 \\ \dot{\omega}_3 \end{bmatrix} = - \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} \times \begin{bmatrix} I_1 \omega_1 \\ I_2 \omega_2 \\ I_3 \omega_3 \end{bmatrix}$$

Closed system

$$I_1 \dot{\omega}_1 = \omega_2 \omega_3 (I_2 - I_3)$$

$$I_2 \dot{\omega}_2 = \omega_1 \omega_3 (I_3 - I_1)$$

$$I_3 \dot{\omega}_3 = \omega_1 \omega_2 (I_1 - I_2)$$

$$\begin{aligned} \dot{\omega}_1 &= \frac{(I_3 - I_1)}{I_1} \omega_2 \omega_3 \\ \dot{\omega}_2 &= \frac{(I_3 - I_1)}{I_2} \omega_1 \omega_3 \\ \dot{\omega}_3 &= \frac{(I_1 - I_2)}{I_3} \omega_1 \omega_2 \end{aligned}$$

Non-linear ODE

Any solution:

$$\omega_1 = \omega_2 = \omega_3 = 0$$

$$\omega_B = \begin{bmatrix} \omega_1^* \\ 0 \\ 0 \end{bmatrix} \sim \begin{bmatrix} 0 \\ \omega_2^* \\ 0 \end{bmatrix} \sim \begin{bmatrix} 0 \\ 0 \\ \omega_3^* \end{bmatrix} = \text{constant}$$

Are these stable?

Linearize near $\omega = \begin{bmatrix} \omega_1^* \\ 0 \\ 0 \end{bmatrix}$

$\omega = \begin{bmatrix} \omega_1^* + \varepsilon_1 \\ 0 + \varepsilon_2 \\ 0 + \varepsilon_3 \end{bmatrix}$ Add small perturbation

$$\dot{\varepsilon}_1 = \frac{(I_2 - I_3)}{I_1} \varepsilon_2 \varepsilon_3 \quad \varepsilon_i \ll \omega_1^*$$

$$\dot{\varepsilon}_2 = \frac{(I_3 - I_1)}{I_2} (\omega_1^* + \varepsilon_1) \varepsilon_3$$

$$\dot{\varepsilon}_3 = \frac{(I_1 - I_2)}{I_3} (\omega_1^* + \varepsilon_1) \varepsilon_2$$

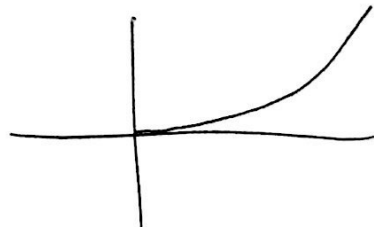
Neglect higher order terms

$$\begin{cases} \dot{\varepsilon}_1 = 0 \\ \textcircled{1} \dot{\varepsilon}_2 = c_2 \varepsilon_3 \\ \textcircled{2} \dot{\varepsilon}_3 = c_3 \varepsilon_2 \end{cases} \quad \begin{aligned} c_2 &= \frac{\omega_1^*}{I_2} (I_3 - I_1) \\ c_3 &= \frac{\omega_1^*}{I_3} (I_1 - I_2) \end{aligned}$$

Linear ODEs

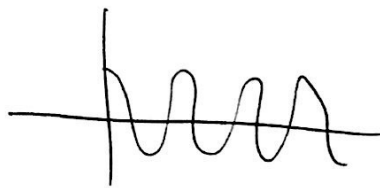
$$\frac{d}{dt} \textcircled{1} + \text{sub m} \quad \textcircled{2} \rightarrow \boxed{\dot{\varepsilon}_2 = c_2 \dot{\varepsilon}_3 = c_2 \varepsilon_3 \varepsilon_2}$$

$$c_2 c_3 > 0 \quad \varepsilon_2 = e^{(\quad)t}$$



diverges

$$c_2 c_3 < 0 \quad \varepsilon_2 = \sin(\quad)t$$



$$C_2 C_3 = \frac{\omega_1^2}{I_2 I_3} (I_3 - I_1)(I_1 - I_2)$$

stable	$\left[\begin{array}{l} < 0 \\ > 0 \end{array} \right.$	$I_1 > I_2$ & $I_1 > I_3$	OR	$I_1 < I_2$ & $I_1 < I_3$
unstable		$I_3 > I_1 > I_2$	OR	$I_2 > I_1 > I_3$

Intermediate axis

Rotation about the intermediate axis is unstable